

2. Shepley L. Ross: *Differential Equations*, John Wiley & Sons, 1984.
3. M. D. Raisinghania: *Ordinary & Partial Differential Equation*, S. Chand & Co. Ltd, 2017.
4. B. Rai, D. P. Choudhary and H. J. Freedman: *A Course of Ordinary Differential Equations*, Narosa, 2002.

Further Readings:

1. Earl A. Coddington and Norman Levinson: *Theory of Ordinary Differential Equations*, McGraw-Hill Edition, 1998.
2. Ravi P. Agarwal and Donal O'Regan: *Ordinary and Partial Differential Equations*, Springer, 2009.
3. Martin Braun: *Differential Equations and Their Applications*, Springer, 1993.
4. Erwin Kreyszig: *Advanced Engineering Mathematics*, John Wiley & Sons, 2011.
5. Ian N. Snedden: *Elements of Partial Differential Equations*, Dover Publication, 2013.
6. Digital Platform: NPTEL/SWAYAM/MOOCs.

Course Prerequisites: To study this course, a student must have studied the Mathematics of class 12th standard and completed mathematics courses of I, II and III semesters.

Year	Semester	Course Code	Paper Title	Theory/Practical	Credits
Diploma in Science (Mathematics as one of the major Subject)					
3	V		Abstract Algebra	Theory	(5+0+2)=6
			Linear Algebra	Theory	(5+0+2)=6
3	VI		Analysis	Theory	(5+0+2)=6
			Numerical Analysis	Theory	(5+0+2)=6

Semester-V

Paper-I (Theory)

Course Title: Abstract Algebra

Programme/Class: Bachelor of Science	Year: Third	Semester: Fifth
Paper-1 Theory Subject: Mathematics		
Course Code:	Course Title: Abstract Algebra	

Course Outcomes: This course is useful to understand the concepts of algebraic structures and their properties. It will help the students for better understanding of other subjects, especially atomic structures in chemistry and certain concepts of physics.

Credits: 6	Compulsory
Max. Marks: 25+75	Minimum Passing Marks:

Total Number of Hours = 70-75

Unit	Contents	Number of
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		Hours
Unit I	Groups: Binary operation and Algebraic structure, Subgroups, Permutation groups, Cyclic groups, Coset decomposition, Lagrange theorem and its consequences, Normal subgroups, Quotient group.	10-15
Unit II	Homomorphism and Isomorphism, Fundamental theorems of homomorphism, Cayley's theorem, Automorphism and inner automorphism, Automorphism groups and their computation, Normalizer and center of group, Finite groups, Commutator subgroups.	12-15
Unit III	Direct Product, Group actions, Stabilizers and orbits, Conjugacy classes, Cauchy Theorem, Simple groups, Sylow's Theorems and their applications	10-15
Unit IV	Rings, Sub rings, Integral domain, Field, Skew field, Ideals, Characteristic of a ring, Ring Homomorphism, Quotient rings, Principal ideals, Maximal ideals, Prime ideals, Principal ideal domains, Polynomial rings and irreducibility.	10-15
Unit V	Field of quotients of an integral domain, Embedding of an integral domain in a field, Factorization in an integral domain, Divisibility, Units, Associates, Prime and irreducible elements, Unique Factorization Domain, Euclidean rings.	12-15

Books recommended:

1. I. N. Herstein: *Topics in Algebra*, John Wiley & Sons, 2006.
2. Joseph A. Gallian: *Contemporary Abstract Algebra*, Narosa Publishing House, 2016.
3. David S. Dummit and Richard M. Foote: *Abstract Algebra*, John Wiley & Sons, 2011.
4. Surjeet Singh and Qazi Zameeruddin: *Modern Algebra*, Vikas Publishing House, India, 2021.

Further Readings:

1. Michael Artin: *Algebra*, Pearson Education, 2015.
2. N. Jacobson: *Lectures in Abstract Algebra-Vol. I, II & III*, Springer, 2013.
3. N. Jacobson: *Basic Algebra-Vol. I & II*, Dover Publications Inc., 2009.
4. R. S. Aggarwal: *A Textbook on Modern Algebra*, S Chand & Company, 1973.
5. Digital Platform: NPTEL/SWAYAM/MOOCs.

Course Prerequisites: To study this course, a student must have studied mathematics courses of I, II and III semesters.

Semester-V
Paper-II (Theory)
Course Title: Linear Algebra

Programme/Class: Bachelor of Science	Year: Third	Semester: Fifth
Paper-II Theory Subject: Mathematics		
Course Code:	Course Title: Linear Algebra	

Course Outcomes: Upon successful completion of this course, the students will be able to understand the theory used to solve the mathematical problems. It also helps to enhance the critical thinking of the students.

Credits: 6	Compulsory
Max. Marks: 25+75	Minimum Passing Marks:

Total Number of Hours = 70-75

Unit	Contents	Number of Hours
Unit I	Vector space, subspaces, Linear combinations, linear spans, Sums and direct sums, Linear dependence and independence, Bases and dimensions, Dimensions and subspaces, Coordinates and change of bases.	10-15
Unit II	Linear transformations, rank-nullity theorem, Linear operators, Invertible linear transformations, Matrix representation of a linear transformation, Transpose of a linear transformation, Similarity of Matrices, Linear functional, Dual space and dual basis, Second dual space, hyperspace.	12-15
Unit III	Eigen values and Eigen vectors, Algebraic and Geometrical Multiplicity, Characteristic and Minimal Polynomials, Annihilators, Cayley-Hamilton theorem, Similar Matrices, Diagonalizable operator.	10-15
Unit IV	Invariant Subspaces, Direct sum decomposition, Projection on a vector space, Primary decomposition theorem, Canonical Forms, Diagonal forms, Triangular forms, Jordan forms.	10-15
Unit V	Inner Product Space, Gram Schmidt orthogonalization Process, Orthogonal Complements. Quadratic Forms, Congruence of Matrices, Reduction and Classification of a real quadratic form, Canonical and Normal form of a real quadratic form, Rank, Signature and Index, Various classes of a real quadratic form.	12-15

Books Recommended:

1. K. Hoffman and R. Kunze: *Linear Algebra*, Prentice Hall of India, 1972.
2. K. B. Dutta: *Matrix and Linear Algebra*, Prentice Hall of India, 2004.
3. Seymour Lipschutz and Marc L. Lipson: *Linear Algebra*, Schaum's Outline Series, McGraw Hill Edition, 2017.
4. S. H. Friedberg, A. J. Insel and L. E. Spence: *Linear Algebra*, Pearson Education, 2015.

Further Readings:

1. G. Hadley: *Linear Algebra*, Narosa Publishing House, 2002.
2. H. Helson: *Linear Algebra*, Hindustan Book Agency, New Delhi, 1994.
3. Gilbert Strang: *Linear Algebra and its Applications*, Cengage Learning India, 2005.
4. Digital Platform: NPTEL/SWAYAM/MOOCs.

Course Prerequisites: To study this course, a student must have studied mathematics courses of I, II and III semesters.

Semester-VI